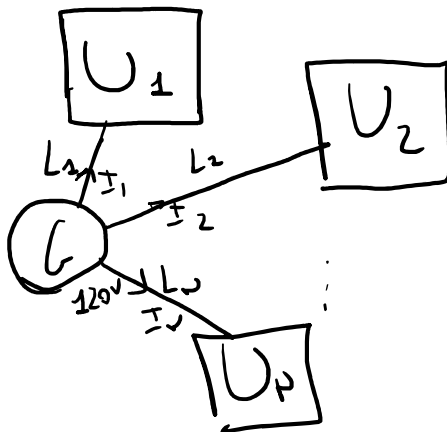


RÉGIME SINUSOÏDALE

mercoledì 20 aprile 2022 16:13

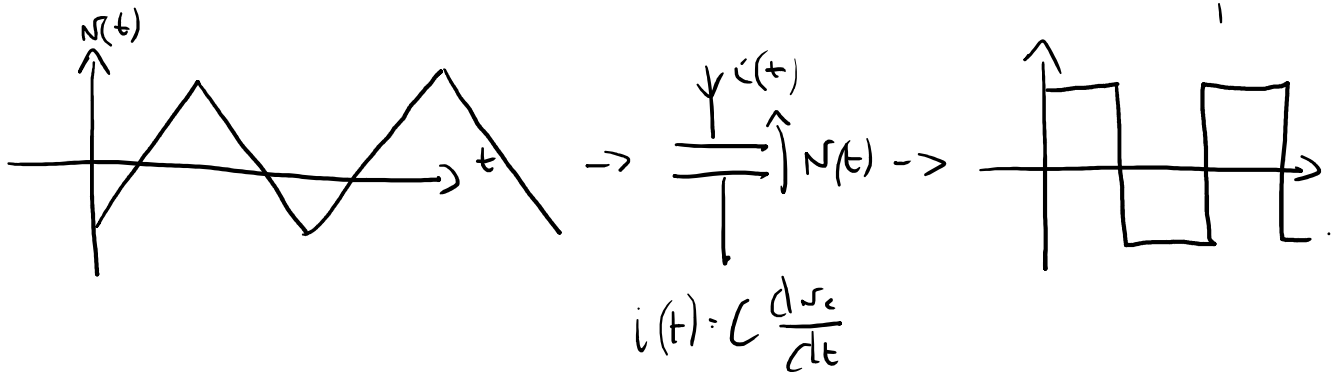
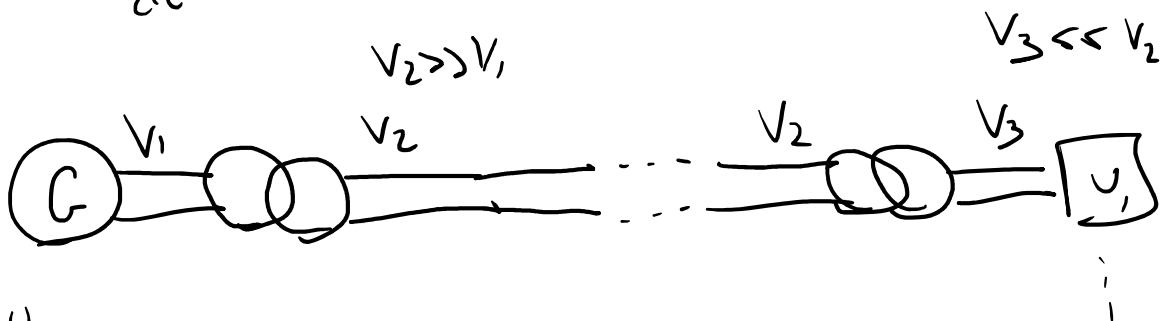


$$I_1 = \frac{P_2}{V_n} \quad \uparrow \quad P_1 \uparrow \rightarrow I_1 \uparrow$$

$$P_{diss} = R \cdot I^2$$

$\downarrow$
 $\frac{R}{S}$

$$f_{em} = - \frac{d\Phi(t)}{dt}$$



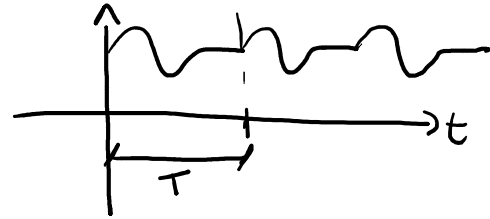
SEGNALI PERIODICI

mercoledì 20 aprile 2022 17:03

Una grandezza temporale $a(t)$ si definisce periodica quando si ripete ad ogni periodo T : $a(t)$

$$a(t) = a(t + nT) \quad \forall t$$

PERIODO
↑
numero
intero

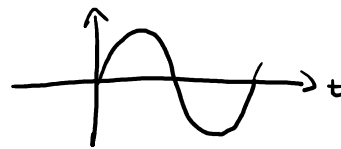
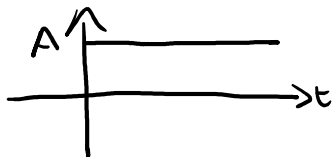


T : PERIODO [s]

f : FREQUENZA [$1/s$] [Hz] = $\frac{1}{T}$ NUMERO DI PERIODI NELL'UNITÀ DI TEMPO

A_m : VALORE MEDIO: $A_m = \frac{1}{T} \int_0^T a(t) dt$

A_{eff} : VALORE EFFICACE: $A_{eff} = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt}$



SEGNALI PERIODICI



SEGNALI ALTERNATI ($A_m = 0$) VALORE MEDIO Nullo



SEGNALI SINUSOIDALI

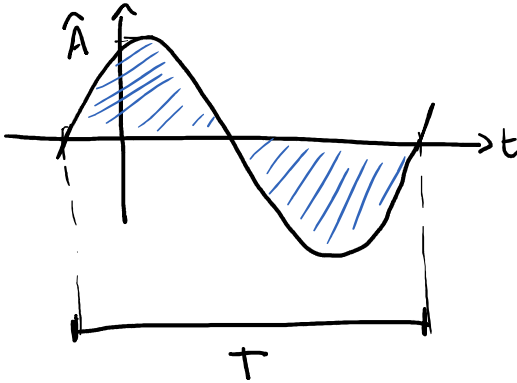
SEGNALI SINUSOIDALI

mercoledì 20 aprile 2022 17:16

$$q(t) = \hat{A} \cos(\omega t + \alpha)$$

↑ VALORE DI PICCO
↑ PULSATIONE
↑ FASE

$$\cos(x) = \sin(x + \frac{\pi}{2})$$



$$A_m = \text{val. medio} = \frac{1}{T} \int_0^T \hat{A} \cos(\omega t + \alpha) dt = 0$$

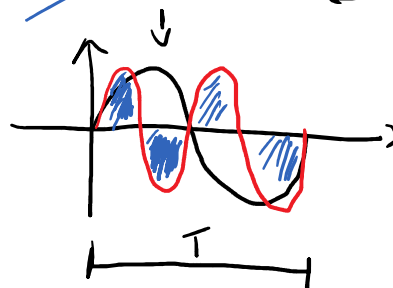
$$A = \text{val. eff.} = \sqrt{\frac{1}{T} \int_0^T [\hat{A} \cos(\omega t + \alpha)]^2 dt} =$$

$$= \sqrt{\frac{1}{T} \int_0^T \hat{A}^2 \cos^2(\omega t + \alpha) dt} = \sqrt{\frac{\hat{A}^2}{T} \int_0^T \cos^2(\omega t + \alpha) dt} =$$

FORMULA DI DUPLIC.
 $\cos^2(x) = \frac{1 + \cos(2x)}{2}$

$$= \sqrt{\frac{\hat{A}^2}{T} \int_0^T \frac{1 + \cos(2\omega t + 2\alpha)}{2} dt} =$$

$$= \sqrt{\frac{\hat{A}^2}{T} \left[\int_0^T \frac{1}{2} dt + \frac{1}{2} \int_0^T \cos(2\omega t + 2\alpha) dt \right]}$$



$$= \sqrt{\frac{\hat{A}^2}{T} \left(\frac{1}{2} T \right)} = \frac{\hat{A}}{\sqrt{2}} \Rightarrow A = \frac{\hat{A}}{\sqrt{2}}$$

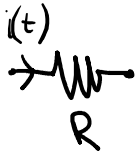
↑
VAL. EFF.

→ FATTORI DI FORMA

SEMPRE USATO
PER UNA SINUSOIDE

POTENZA MEDIA DISSIPATA DA R

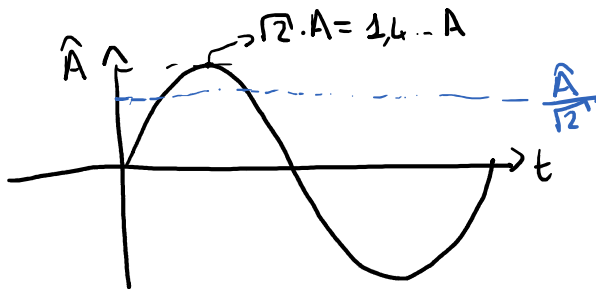
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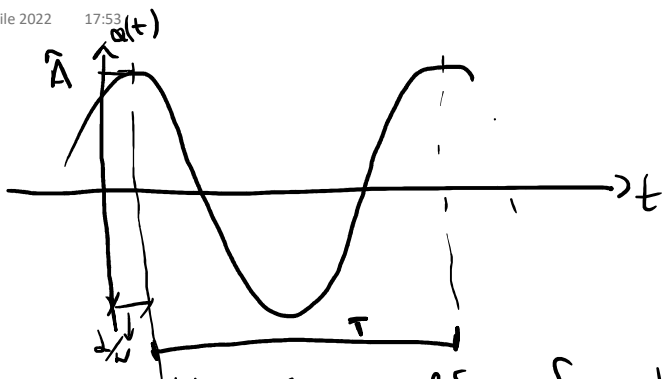
$$P(t) = R i^2(t)$$

$$i(t) = \hat{I} \cos(\omega t + \alpha)$$

$$\langle P(t) \rangle = \frac{1}{T} \int_0^T R i^2(t) dt = R \underbrace{\frac{1}{T} \int_0^T i^2(t) dt}_{I^2} = R I^2$$



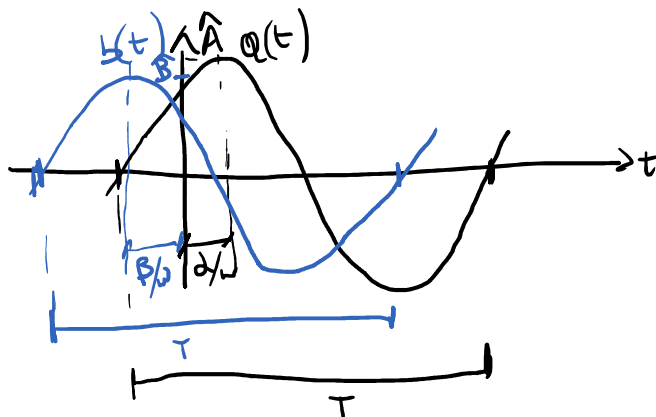
Una corrente DC di 1A compie una potenza dissipata sul resistore pari a quella di una corrente sinusoidale di valore efficace pari a 1A



Una sinusoide è completamente definita da 3 parametri:

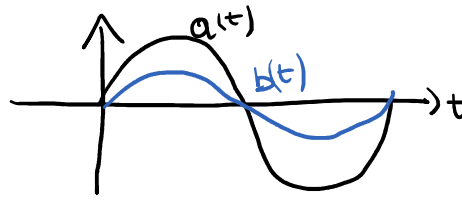
- Valore massimo $\hat{A}$ o val. efficace A : $A = \frac{\hat{A}}{\sqrt{2}}$
- pulsazione ω o frequenza f o periodo T : $f = \frac{1}{T}$; $\omega = 2\pi f$
- fase α

$$\left. \begin{aligned} q(t) &= \hat{A} \cos(\omega t + \alpha) \\ b(t) &= \hat{B} \cos(\omega t + \beta) \end{aligned} \right\} \begin{array}{l} \text{SEGNALI} \\ \text{ISOFREQUENZIALI} \end{array}$$

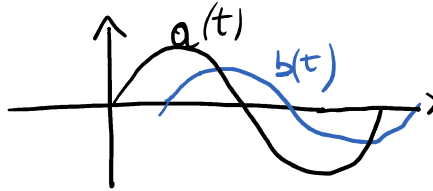


$$\varphi = \text{sfasamento} = \alpha - \beta \quad (\text{DIFFERENZA DI FASE})$$

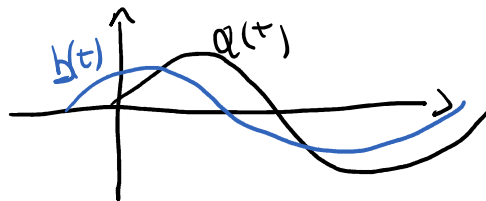
$\varphi = 0$: $a(t)$ e $b(t)$ sono in FASE



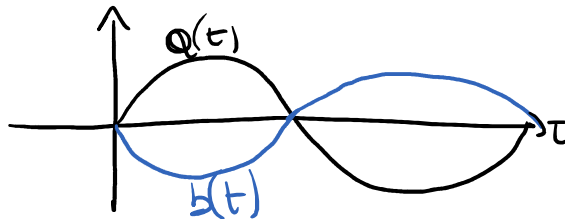
$\varphi > 0$: $a(t)$ è in ANTICIPO rispetto a $b(t)$
 $b(t)$ è in RITARDO rispetto a $a(t)$



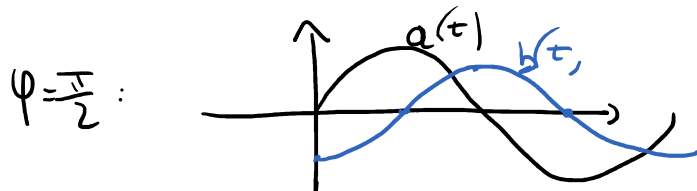
$\varphi < 0$: $a(t)$ è in RITARDO rispetto a $b(t)$
 $b(t)$ è in ANTICIPO rispetto a $a(t)$



$\varphi = \pm \pi$: $a(t)$ e $b(t)$ sono in OPPOSIZIONE DI FASE



$\varphi = \pm \frac{\pi}{2}$: $a(t)$ e $b(t)$ sono in QUADRATURA



$\varphi = \frac{\pi}{2}$:

$b(t)$ è in
QUADRATURA in
RITARDO